PROPOSAL FOR LRDE · BENGALURU · RESTRICTED

Altera

AI Compute Platform for LRDE

Sovereign training hardware and Altera Agilex™ FPGA inference for radar signal intelligence, AESA processing aids, and real-time classification.

Electronics & Radar Development Establishment | Training + Inference | Agilex FPGA AI Suite

AUDIENCE
LRDE leadership & radar AI teams

FOCUS
Radar / AESA / EW-adjacent AI

VERSION
1.0 · Aug 2026

AGENDA

What we will cover

A complete path from LRDE training data to fielded FPGA inference inside radar timelines.

1. LRDE mission fit & AI opportunities
2. End-to-end architecture
3. Training hardware (secure GPU lab)
4. Inference on Altera Agilex FPGAs
5. Software toolchain & MLOps
6. LRDE use-case packages
7. Security & sovereignty
8. PoC roadmap & next steps

LRDE CONTEXT

Radar AI where it matters

LRDE designs ground-based, shipborne, airborne, and space-related radar systems — AI must respect real-time DSP pipelines and SWaP.

AESA & multifunction

Fire-control, surveillance, 4D radars — multi-mode interleaved operation needs deterministic assistive AI.

Signal intelligence

Pulse / emitter classification, LPI detection aids, clutter/anomaly separation on I/Q & spectrograms.

Advanced research

Hypersonic cues, stealth-detection aids, FOPEN features, netted-radar fusion — train secure, infer at the sensor.

PROBLEM

Gaps in today's radar AI path

Training

- Classified I/Q & track data cannot leave LRDE
- Cloud GPUs unsuitable for air-gapped labs
- Weak experiment / model lineage

Inference

- GPUs struggle with radar SWaP & latency budgets
- Need tight coupling to JESD / ADC / beam data
- Long program life vs short GPU cycles

Required outcome

A sovereign stack that lets LRDE:

- **Train** on-prem on radar datasets
- **Compile** models to FPGA AI IP
- **Deploy** inside ERP / processor cabinets or payloads
- **Update** bitstreams without board respins

ARCHITECTURE

Train once · Infer on Agilex

```
LRDE SECURE LAB
  Radar Data Vault → GPU Training Cluster → Model Registry
                        │  ONNX / OpenVINO IR
                        ▼

  Quantize + Accuracy Gates → FPGA AI Suite (dla_compiler)
                        │
          ┌─────────────┼──────────────┐
          ▼             ▼              ▼
   Hostless Agilex   SoC Agilex 5    PCIe Agilex 7
   (ERP / payload)   (edge node)     (shelter / C2)
              ALTERA FPGA INFERENCE TIER
```

GPU

Secure training & fine-tune

OpenVINO

IR + hetero runtime

Agilex

Deterministic field inference

TRAINING HARDWARE

Air-gapped radar AI training lab

Sized for LRDE model development on spectrograms, RD maps, tracks, and multi-sensor fusion sets.

COMPONENT	REFERENCE SPEC	ROLE FOR LRDE
GPU nodes	8× datacenter GPUs/node, dual CPU, 1–2 TB RAM	CNN / transformer training
Fabric	NVLink + InfiniBand NDR / RoCE	Distributed training
Storage	Encrypted 0.5–2 PB object / parallel FS	I/Q archives, checkpoints
CPU ETL	High-core labeling & feature nodes	RD maps, CFAR labels, sim data
Security	Air gap, bastion MFA, optional diode	Sovereign model export only

TRAINING SOFTWARE

ML stack inside the wire

Frameworks & ops

- PyTorch / TensorFlow (air-gapped registry)
- DeepSpeed / FSDP for large models
- Slurm or Kubernetes + Apptainer
- MLflow / DVC for lineage

Radar-specific data path

- I/Q → STFT / RD / spectrogram pipelines
- Synthetic + trial data blending
- Golden sets for accuracy gates
- Signed ONNX / IR promotion

→

Handoff to inference

Export ONNX → OpenVINO IR → FPGA AI Suite compile → signed bitstream / network-bin for LRDE platforms.

INFERENCE HARDWARE

Altera Agilex™ for radar timelines

FPGAs sit next to the exciter–receiver–processor chain — reconfigurable AI without breaking determinism.

DEVICE	INT8 CLASS*	LRDE PLACEMENT
Agilex 3	~2.6 TOPS	Miniaturized / SoC radar sensors
Agilex 5 E-Series	~26 TOPS	Compact processors, mobile radars
Agilex 5 D-Series	~152 TOPS	Multi-channel classification
Agilex 7 I/M-Series	~89 TOPS class	ERP co-process, PCIe, shelters

*Architecture- and IP-dependent; sized per model with FPGA AI Suite Architecture Optimizer.

DEPLOYMENT

Form factors for LRDE systems

Hostless / DDR-free

AI IP in fabric beside DSP. Lowest latency for pulse/window classifiers inside ERP.

SoC (HPS)

On-chip Linux + AI IP. Ideal for transportable radars and edge fusion nodes.

PCIe / OFS card

Look-aside acceleration in operation shelters and lab validation racks.

JESD204 / ADC attach

Aurora / Ethernet sensor nets

Bitstream encryption

Long defence lifecycle

WHY ALTERA

FPGA vs GPU for LRDE inference

CRITERION	GPU	ALTERA FPGA
Latency determinism	Batch / driver variable	Pipeline-friendly, stable
RF / DSP coupling	Needs host staging	Direct fabric I/O
Power in cabinet / payload	High	Lower W for fixed nets
Model update	Driver + app stack	Bitstream / network-bin
Program longevity	Short SKU cycles	10–15+ year class support

SOFTWARE TOOLCHAIN

From PyTorch to radar bitstream

```
Train (PyTorch) → ONNX
  → OpenVINO convert_model / ovc  →  IR (.xml/.bin)
  → Quantize (INT8 / FP16) + golden accuracy gate
  → FPGA AI Suite  dla_compiler --march <arch.arch>
  → Architecture Optimizer (area ↔ FPS)
  → Bit-accurate OpenVINO emulation
  → Quartus Prime integrate AI IP → bitstream
  → Runtime: FPGA  or  HETERO:FPGA,CPU  (or hostless)
```

Core tools: OpenVINO · FPGA AI Suite · Quartus Prime · optional Open FPGA Stack (OFS).

Three PoC-ready use cases

1. Emitter / pulse ID

Spectrogram CNN on Agilex 7 hostless or PCIe. Continuous windows, deterministic latency for EW-adjacent radar modes.

2. Target / clutter assist

RD-map or tracklet classifier on Agilex 5. Aids discrimination for fighters, slow movers, false-alarm reduction.

3. Netted fusion node

Multi-radar feature fusion on Agilex 7 PCIe in shelter C2 — signed model hot-swap after trials.

SECURITY

Sovereignty by design

Data residency

I/Q, tracks, and labels never leave LRDE premises.

Air gap

Training isolated; controlled diode export of IR/bitstreams only.

Bitstream trust

Encryption, authentication, anti-tamper options on Agilex.

Indian partners

Board / SOM qualification with SI & DPSU ecosystem.

ROADMAP

Phased engagement with LRDE

0 **Discovery — 4–6 weeks**
Workload survey (2 models), latency/SWaP envelopes, success metrics with ERP owners.

1 **PoC — 8–12 weeks**
Port models to Agilex eval kits; accuracy vs golden set; latency report for radar windows.

2 **Pilot — 3–6 months**
Rugged module or PCIe in processor cabinet; training-cluster slice; MLOps baseline.

3 **Scale — program-driven**
Production SOM/PCIe, environmental qualification, ILS & field update process.

REFERENCE BOM

What LRDE would procure

Training lab

- GPU nodes + IB fabric
- Encrypted storage
- Hardened MLOps stack

Inference & tools

- Agilex 7 PCIe + Agilex 5 kits
- Quartus Prime + FPGA AI Suite
- OpenVINO + enablement services

Detailed commercial annex after discovery workshop. See `BOM_Reference.md`.

CALL TO ACTION

Recommended next steps

1. Workshop

2-day architecture session at LRDE — nominate radar AI & ERP stakeholders.

2. Select models

Pick two flagship networks (e.g. pulse ID + RD classifier) for Phase-1 PoC.

3. Lock gates

Agree accuracy, window latency, and power budgets before compile.

4. SOW

Issue technical SOW + commercial annex for PoC kits and FAE support.

PREPARED FOR
LRDE, DRDO Bengaluru

INFERENCE
Altera Agilex + FPGA AI Suite

DOCS
proposal/LRDE_AI_Slide_Deck.html